

# An alternative mechanism for recent volcanism on Mars

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[1] Images of the Martian surface indicate that volcanism is a recent process on Mars with ages of the youngest lava flows of about 2 Ma. Interestingly, these young lava flows are confined to the large, old volcanic centers of Tharsis and Elysium, which exhibit a thickened crust compared to surrounding areas. The origin of the volcanic centers but also their longstanding volcanic activity is often explained by strong mantle plumes arising from the core-mantle boundary. However, their existence and sustainment during the last few billion years is uncertain under Martian conditions. Alternatively, the thermal insulation by the crust could cause the longstanding volcanic activity. We show that a locally thickened crust, which has a lower thermal conductivity and is enriched in radioactive heat sources in comparison to the mantle, leads to significant lateral temperature variations in the upper mantle sufficient to generate partial melt even in the present Martian mantle.

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## 1. Introduction

[2] High resolution images of Martian volcanoes, e.g., by Mars Express have shown that some of them exhibit nearly uncratered and, therefore, extremely young lava flows. Analysis by *Neukum et al.* [2004] leads to the conclusion that some of these flows are as young as 2 Ma. Although the evidence for recent volcanism is overwhelming, the underlying mechanism which causes this ongoing volcanic activity is still not understood. Recent thermal evolution models indicate that volcanism may have stopped at least some hundred millions years ago when the Martian interior cooled significantly below the mantle solidus [*Hauck and Phillips*, 2002; *Breuer and Spohn*, 2003, 2006]. It has been debated if the volcanism of large volcanic centers like Tharsis and Elysium could have been sustained by a strong mantle plume [e.g., *Kieffer*, 2003], bringing enough heat from the core-mantle boundary to the upper mantle to cause melting and volcanism within the stagnant lid. Although this hypothesis may be true for early Mars, a mantle plume – if existent at all – is difficult to sustain under recent conditions and would yield just a slight temperature increase of a few tens of K at the base of the stagnant lid (see discussion in *Schumacher and Breuer* [2006a]). We therefore concentrate on an alternative explanation which does

not involve temperature heterogeneities caused by mantle plumes.

[3] Most thermal evolution models assume a constant thermal conductivity in mantle and crust of  $4 \text{ Wm}^{-1}\text{K}^{-1}$ . In a recent study, however, we have shown that different thermal conductivities in crust and mantle due to compositional, temperature and pressure variations should not be neglected [*Schumacher and Breuer*, 2006a, 2006b]. The value of the crustal conductivity for basaltic material is about  $2 \text{ Wm}^{-1}\text{K}^{-1}$  [e.g., *Clifford and Fanale*, 1985], and the temperature- and pressure-dependent thermal conductivity  $k(T, P)$  in the mantle [*Hofmeister*, 1999] is also lower than the earlier used constant value of  $4 \text{ Wm}^{-1}\text{K}^{-1}$ . As a consequence, the planetary interior – in particular the upper mantle – cools slower: a global partial melt zone at the base of the stagnant lid exists during a substantial period of time and the present temperature of the average mantle is only a few tens of K below the mantle solidus [*Schumacher and Breuer*, 2006a]. Thus, only small lateral temperature variations are needed to generate partial melt. In contrast, for models with a crustal thermal conductivity larger than  $2 \text{ Wm}^{-1}\text{K}^{-1}$  a global partial melt zone disappears already after about one billion years and the present average mantle temperature is more than 100 K below the mantle solidus [see *Schumacher and Breuer*, 2006b]. Partial melting in the present mantle but also during the last few billion years would therefore require large lateral temperature variations.

[4] In the present study we investigate the effects of lateral variations in crustal thickness and of the thermal conductivity in crust and mantle on the temperature distribution of the stagnant lid. In particular we try to understand how much the crustal root of about 20 km underneath the large Tharsis rise [*Montési and Zuber*, 2003] influences the upper mantle temperatures and if it can cause the existence of a partial melt zone.

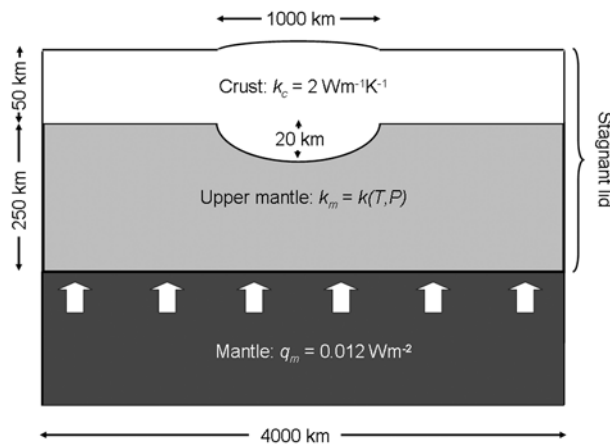
## 2. Models

[5] We solve a two dimensional heat conduction equation with a 2-D finite elements code (COMSOL Multiphysics) for the temperature distribution in crust and mantle for present-day conditions. Boundary conditions are a constant surface temperature, a constant mantle heat flow (whose value is given by 1-D parameterized mantle convection models) as well as thermal insulation for the left and right boundary. The width of the models is chosen in a way that the temperature distribution close to the sideways boundaries is not influenced by the thickened crust, i.e., temperatures there are the same for models with and without increased crustal thickness.

[6] Models with constant crustal thickness consist of a 4000 km wide and 50 km thick crust which is underlain by a 250 km thick conducting upper mantle. We limit the depth of the model to 300 km, which is the thickness of the

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**Figure 1.** Schematic model setup.

stagnant lid obtained from parameterized convection models [Schumacher and Breuer, 2006a]. For models with locally increased crustal thickness, we add two ellipsoids whose semi-major axis' are parallel to the upper and lower crustal boundary, representing the topography and the crustal root (Figure 1). Using ellipsoids instead of rectangular blocks ensures a gradual transition from average to thickened crust. The area of thickened crust has a diameter of 1000 km and exhibits a crustal root of up to 20 km compared to surrounding areas, thereby roughly reproducing the structure of the part of the Tharsis rise, which exhibits the highest crustal thickening (Syria Planum and adjacent areas).

[7] For our standard model, we use present-day thermal conditions derived from thermal evolution models [Schumacher and Breuer, 2006a]: a constant mantle heat flux  $q_m$  of  $0.012 \text{ W m}^{-2}$  and a concentration of radioactive heat sources in the present primitive mantle of  $1.378 \times 10^{-8} \text{ W m}^{-3}$  which is in agreement with estimates from concentrations of incompatible elements in SNC meteorites [e.g., Treiman *et al.*, 1986]. The crust is homogeneously enriched in radioactive elements compared to the primitive mantle by an enrichment factor,  $\Lambda$ . For the Martian crust the enrichment factor is assumed to vary between 5 assuming a basaltic crust and 10 for an andesitic composition [Nimmo and Tanaka, 2005]. The surface temperature is 220 K. We assume a crustal density of  $2900 \text{ kg m}^{-3}$  and a mantle density of  $3500 \text{ kg m}^{-3}$  [Zuber, 2001; Sohl *et al.*, 2005]. The thermal conductivity of the crust,  $k_c$ , is  $2 \text{ W m}^{-1} \text{ K}^{-1}$ , while the mantle thermal conductivity  $k_m$  is temperature- and pressure-dependent [Hofmeister, 1999]. In additional calculations, we have varied the enrichment of crustal heat sources, the crustal thermal conductivity, the thickness of the stagnant lid, the thickness of the average crust and of the crustal root, the mantle heat flow, as well as the diameter of the thickened crustal region. For all parameters used in the calculations see Table 1.

### 3. Results

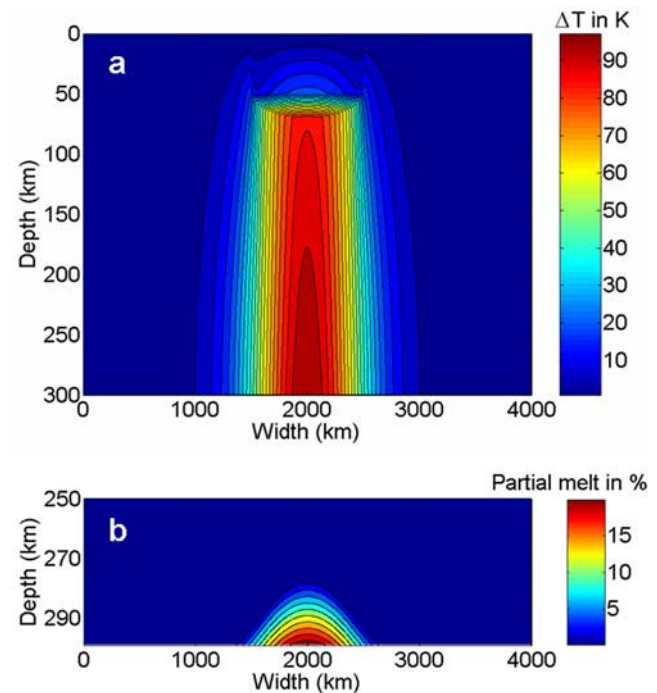
[8] Figure 2 shows the temperature difference,  $\Delta T$ , between models with and without locally increased crustal thickness. A significant temperature increase below regions with thickened crust can be observed with a maximum value close to 100 K. The result shows that lateral heat

**Table 1.** Definition of Parameters and Their Values Used in the Calculations

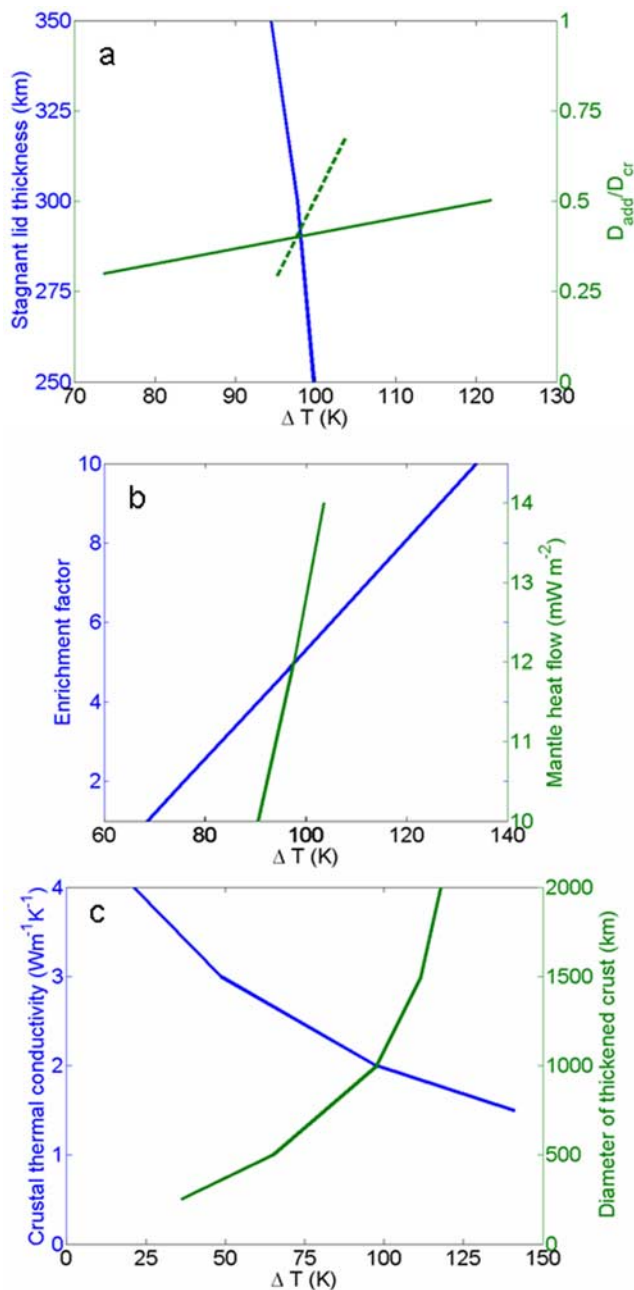
| Parameter                    | Notation  | Value                  | Unit                             |
|------------------------------|-----------|------------------------|----------------------------------|
| Crustal density              | $\rho_c$  | 2900                   | $\text{kg m}^{-3}$               |
| Mantle density               | $\rho_m$  | 3500                   | $\text{kg m}^{-3}$               |
| Mantle heat flow             | $q_m$     | 10, 12 or 14           | $\text{W m}^{-2}$                |
| Mantle thermal conductivity  | $k_m$     | $k(T, P)$              | $\text{W m}^{-1} \text{ K}^{-1}$ |
| Crustal thermal conductivity | $k_c$     | 1.5, 2, 3 or 4         | $\text{W m}^{-1} \text{ K}^{-1}$ |
| Stagnant lid thickness       | $d_{st}$  | 250, 300 or 350        | km                               |
| Average crustal thickness    | $d_c$     | 30, 50 or 70           | km                               |
| Thickness of crustal root    | $d_{add}$ | 15, 20 or 25           | km                               |
| Diameter of crustal root     | $d_{vol}$ | 250, 500, 1000 or 2000 | km                               |
| Enrichment factor            | $\Lambda$ | 1, 2, 5 or 10          |                                  |

transport can not compensate the less efficient heat transport through the crustal root, i.e., temperatures in and below the crustal root are higher than in the surrounding regions.

[9] The increase in temperature can result in the generation of partial melt at the base of the stagnant lid if the average mantle temperature is not more than 100 K lower than the solidus. Figure 2b depicts the amount of partial melt which can be generated by  $\Delta T$  for the model of Figure 2a if the average mantle temperature is just below the solidus temperature. For the calculation of the partial melt content the solidus and liquidus of dry peridotite [Takahashi, 1990] have been chosen. It has to be considered that the melt content and the extent of the melt region change with time due to the changing average Martian mantle temperature.



**Figure 2.** (a) Temperature differences between a model with and a model without crustal thickening (absolute temperature increase caused by crustal thickening). (b) Amount of partial melt which is generated by the increase in temperature for Figure 2a assuming an average mantle temperature close to the solidus of dry peridotite [Takahashi, 1990].



**Figure 3.** Maximal lateral temperature variation at the base of the stagnant lid due to a crust with a low thermal conductivity and locally increased thickness. Effect of variations of (a) the stagnant lid thickness (blue line), the additional crustal thickness (green solid line), and the average crustal thickness (green dashed line), (b) the enrichment factor (blue line) and mantle heat flow (green line), and (c) the crustal thermal conductivity (blue line) as well as the diameter of the thickened crust (green line).

[10] In additional calculations, we conducted a parametric study to evaluate the importance of relevant parameters on the calculated maximum  $\Delta T$  at the base of the lithosphere for models with  $k_m = k(T, P)$ . The thickness of the stagnant lid has only a minor influence on  $\Delta T$ . The temperature differences between models with and without crustal root vary by less than 10 K for a variation of the

stagnant lid thickness of 100 km (Figure 3a, blue line). Variations of the crustal thickness, on the other hand, result in more drastic temperature changes. The ratio of the crustal root thickness to the average crustal thickness has been plotted in Figure 3a; we either fix the crustal root thickness (green solid line) or the average crustal thickness (green dashed line). While variations of the average crustal thickness (green dashed line) cause only moderate temperature differences compared to the reference model (about 5 K for a 20 km difference in average crustal thickness), changes of the thickness of the crust root lead to substantial temperature variations (solid green line). An increase of 5 km of the crustal root thickness results in an increase in  $\Delta T$  of approximately 25 K. Thus, minor local crustal variations may have a significant effect on the local temperature distribution.

[11] The influence of the distribution of radioactive elements between mantle and crust (the enrichment factor, i.e., the ratio of crustal heat sources and of heat sources in the primitive mantle) on the temperature distribution is shown in Figure 3b (blue line). With an enrichment factor of 10, a value that is more representative for a  $\text{SiO}_2$  rich material like andesite, the temperature difference at the base of the lithosphere increases to about 130 K. However, if the crust is not enriched ( $\Lambda = 1$ , homogeneous distribution of heat sources between crust and mantle), the temperature increase is still about 70 K. This increase is only due to the lower crustal thermal conductivity. It can, therefore, be concluded that for the standard model with  $\Lambda = 5$  about 70 % of the temperature change is caused by the thermal conductivity alone. Variations in the average mantle heat flux do not cause significant changes in the maximum temperature at the base of the stagnant lid (Figure 3b, green line). The influences of the crustal conductivity and of the diameter of the thickened crust on the temperature distribution are shown in Figure 3c. The larger the crustal thermal conductivity is, the smaller is the temperature difference between a model with constant crustal thickness and a model with a crustal root. For  $k_c = 4 \text{ W m}^{-1} \text{K}^{-1}$  the temperature increase is still 25 K because of the temperature dependence of the mantle thermal conductivity. The diameter of the thickened crust has also a strong influence on the temperature distribution. As can be expected, the temperature difference caused by the thickened crust approaches zero for decreasing diameters. With increasing size of the region with thickened crust,  $\Delta T$  approaches a value that can be calculated if the lateral heat flow is neglected. Interestingly, for smaller diameters the largest temperature difference is located right at the base of the thickened crust and not, as in all other models with  $k_m(T, P)$ , at the base of the stagnant lid.

#### 4. Summary and Discussion

[12] The present results suggest that once a large volcanic region with a thickened crustal root is formed, partial melt in the silicate mantle is generated preferentially beneath that region if the crust has different thermal properties as compared to the mantle, i.e., a lower thermal conductivity and a higher concentration of radioactive heat sources. Although the models shown only depict the temperature distribution for present-day conditions, the mechanism of

crustal insulation also acted in the past as the bulks of Tharsis and Elysium have been built relatively early in the planet's history [e.g., *Anderson et al.*, 2001; *Werner*, 2006]. The results show that strong longstanding stable mantle plumes are not required to explain the ongoing volcanic activity on Mars [e.g., *Kieffer*, 2003] as long as the crust and its distinct physical properties are considered.

[13] We find for our standard model a temperature increase of about 100 K below the volcanic region as compared to the surroundings. This lateral temperature variation is sufficient to produce locally partial melt at the base of the stagnant lid even for present-day thermal conditions. Thermochemical evolution models for Mars assuming the same thermal properties for the crust and the mantle as for the models presented require lateral present-day temperature variations of more than about 50 K to produce partial melt [*Schumacher and Breuer*, 2006b]. In the early evolution even smaller lateral temperature variations have been sufficient to produce partial melt and during the period of time when a global partial melt zone existed in Mars, the melt content beneath the large volcanic regions was significantly enhanced in comparison to the mean melt content.

[14] Melt generation on Mars by the proposed mechanism depends mainly on the low crustal thermal conductivity in comparison to the mantle conductivity. An increase of the crustal conductivity results in a decrease of the lateral temperature variations and, on the other hand, in an increase of the difference between the solidus and the average mantle temperature [*Schumacher and Breuer*, 2006b]. As a consequence, it becomes more and more difficult to generate partial melt for higher crustal thermal conductivities. The thermal conductivity of the majority of compact volcanic materials ranges between 1.5 and 3.5  $\text{Wm}^{-1}\text{K}^{-1}$  at ambient temperatures [e.g., *Clifford and Fanale*, 1985; *Clauser and Huenges*, 1995] but decreases with temperature like for mantle material [e.g., *Seipold*, 1998]. For example, the thermal conductivity of typical compact basalt decreases from about 2  $\text{Wm}^{-1}\text{K}^{-1}$  at 270 K to about 1.5  $\text{Wm}^{-1}\text{K}^{-1}$  at 800 K. Given the range of possible variations in the composition of the compact crust and the influence of the temperature on the thermal conductivity, a column-averaged thermal conductivity of about  $2 \pm 1 \text{ Wm}^{-1}\text{K}^{-1}$  can be assumed for the entire Martian crust [*Clifford and Fanale*, 1985; *Turcotte and Schubert*, 2002]. An insulating effect of the Martian crust as shown in the present study is therefore likely and should not be neglected in thermal evolution models.

[15] An additional layer of dry regolith, which has an even lower thermal conductivity than the crust, would amplify the thermal effect already seen. However, the thermal conductivity of regolith whose pore space is filled by ice or water approximately equals that of solid crust. In this case the regolith would cause no discernable effect. As the vertical extent and composition of the regolith are not known, we do not consider this extra layer in our calculations.

[16] In addition to the insulating effect of the crust, the temperature dependence of the mantle thermal conductivity is an essential factor for the generation of partial melt. For increasing mantle temperatures the thermal conductivity decreases, which results in a further increase of the temperatures. At the base of the stagnant lid, the maximum amount

of partial melt is produced since here the average mantle temperature is also closest to the solidus. Extruded partial melt from that region can be replenished from the convecting mantle, thereby providing the continuation of the melting process. Due to the replenishment of extruded material by the convecting mantle, a significant depletion of the mantle in radioactive elements below the isolating structure, which may counteract the lateral temperature variation, is unlikely.

[17] The lateral temperature variations depend also on the lateral extension of the crustal root. If the lateral extension of the crustal root is smaller than about 500 km in diameter, the temperature variations may become too small to produce melt locally under present thermal condition on Mars. Hence, only large volcanic centers like Tharsis and Elysium are likely to generate partial melt for a substantial period of time during the evolution. The diameter of 1000 km, which is used for the standard model, resembles in size not only the main part of the Tharsis rise but also Elysium. However, in case of Elysium the average crustal thickness (about 35 km) and the crustal root (about 15 km) are smaller [*Montési and Zuber*, 2003]. Additional calculations have shown that for such parameters the temperature increase is about 70 K and thus possibly still sufficient for the generation of partial melt. The temperature increase underneath Tharsis, on the other hand, is probably significantly higher than 100 K as a diameter of 1000 km solely fits to the most central part of the Tharsis rise.

[18] The assumption of a constant mantle heat flow into the stagnant lid is an oversimplification that might alter the results. In an extreme case, heat flow variations might totally compensate the temperature increase at the base of the stagnant lid if the heat flow underneath the crustal root is smaller than in the surrounding regions. We have tested this possibility with 2-D convection models. The results show that the heat flow underneath regions with higher crustal thickness is not diminished, but equal or even slightly enhanced compared to surrounding areas.

[19] The thickness of the stagnant lid limits the depth from which the melt rises toward the surface to a certain degree. Thermochemical evolution models show that the stagnant lid grows during the first 750 Ma to a thickness of about 300 km and changes only marginally from then on [*Schumacher and Breuer*, 2006a]. 750 Ma are also about the time when the main bulge of Tharsis and Elysium might have formed [*Werner*, 2006]. Thus, subsequent volcanic activity suggests that melt must have risen from these deep regions. From simple hydrostatic considerations, the height of Olympus Mons with about 26 km can only be explained with such a deep magma reservoir. Although the mechanism of melt transport is not well understood, there are several factors which promote the idea of longstanding volcanism focused to areas of large volcanic regions, even at times when a global partial melt zone existed [*Schumacher and Breuer*, 2006a]. First of all, the melt content below the large volcanic regions but also the temperatures in crust and upper mantle in these regions are enhanced compared to surrounding areas as shown in Figure 2, making it thereby significantly easier for melt to rise to the surface. Within the crust and elastic part of the mantle a reactivation of old volcanic vents and fissures is likely. Moreover, the concentration of recent volcanism to old volcanic regions is also

avored due to density contrasts between these regions and the average crust. As the density of the Martian average crust is suggested to be relatively low with values of 2600 to 2900 kg m<sup>-3</sup> [McGovern *et al.*, 2004; Lowry and Zhong, 2003] and that of volcanic regions relatively high with values of about 3000 to 3300 kg m<sup>-3</sup> [McKenzie *et al.*, 2002; McGovern *et al.*, 2004; Belleguic *et al.*, 2005], it is much more likely for dense basaltic melt to reach the surface in regions of high density, i.e., old volcanic centers.

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